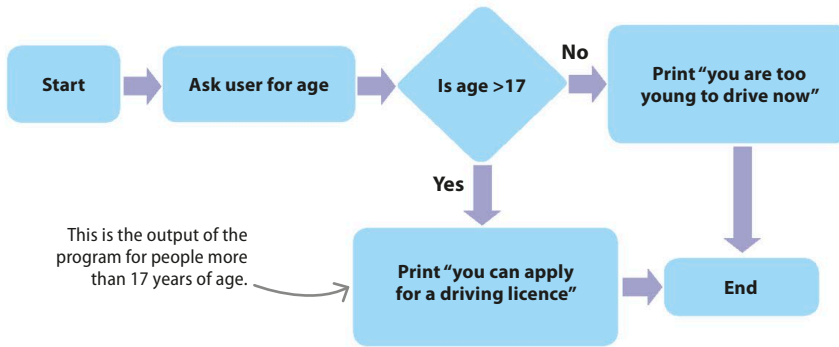


Describing algorithms

Algorithms can be described using flowcharts or pseudocode. A flowchart is made up of boxes linked by arrows. Each box contains a step in solving the problem or a question. Pseudocode is laid out like a computer program, but it's written in a human language.



Ask the user to enter their age in years

If the user's age is greater than 17

print "You can apply for a driving licence"

else

print "You are too young to drive just now"

◁ Pseudocode

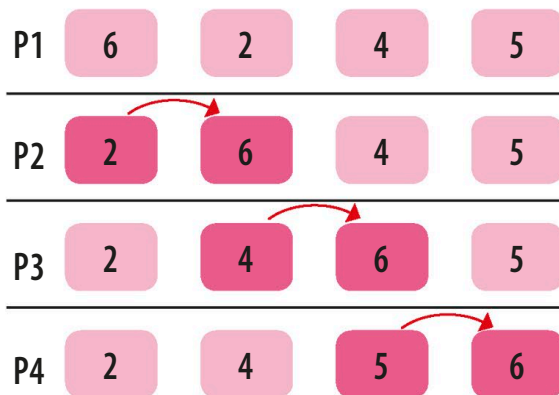
Describing algorithms in pseudocode allows programmers to understand them, no matter what computer languages they are familiar with. This makes the whole range of algorithms available.

BIOGRAPHY

Muhammad Al-Khwārizmī

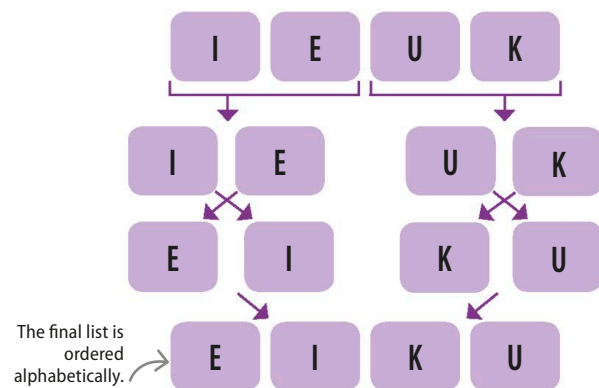


The word "algorithm" comes from the name of 9th-century mathematician Abu Abdullah Muhammad ibn Mūsā al-Khwārizmī. Al-Khwārizmī lived in Baghdad, Iraq, and translated a number of scientific books from ancient Greek and Sanskrit into Arabic. He also wrote several books on mathematics, astronomy, geography, and history. These books were later translated into Latin and studied in European universities. The word "algebra" comes from the title of one of his books.



△ Bubble sort

This looks at the items a pair at a time, swapping them round if the second one of the pair is larger than the first. It's not very efficient as it's often necessary to go through the list several times.



△ Merge sort

This breaks a list of items into many tiny lists. It then merges all these lists into newly sorted ones, finally producing a single sorted list. It uses more code than bubble sort, but is more efficient.